Find what is needed to make the forces balance

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Equilibrium / ES: equilibrio / PT: equilíbrio

Acceleration is zero. The forces sum to zero in every direction.

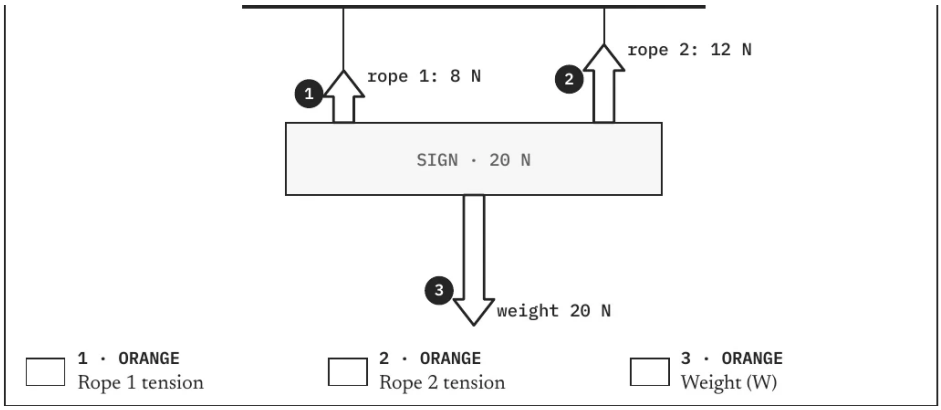
Also on your note page: Static and dynamic equilibrium, Missing force, Instantaneous rest.

B READ THE RELATIONSHIP

An object is in translational equilibrium when its forces add to zero.

Equilibrium requires zero force sum on every axis

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other.

Given weight 40 N rope 1 = 15 N at rest

Rule up = down

Work 15 + T 2 = 40 T 2 = 40 − 15

Answer 25 N up

E SAVE FOR THE FINAL MEETING

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

Read the physics: Equilibrium

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Choose horizontal and vertical axes. Add the forces on each axis. When both sums are zero, acceleration is zero. The object can be at rest or moving steadily.

02 HOW IT WORKS

A hanging sign has downward weight and upward support. If the ropes are vertical, add their tensions. If the ropes are tilted, only the upward parts support the weight.

03 WHAT TO CHECK

Being at rest for an instant is not enough to prove equilibrium. Look for zero net force and unchanged velocity.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other. 25 N up.

05 CONNECT TO XJ

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

SAY IT BEFORE YOU WRITE

Can an object in equilibrium be moving? The ___ forces balance. Its acceleration is ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Equilibrium / ES: equilibrio / PT: equilíbrio

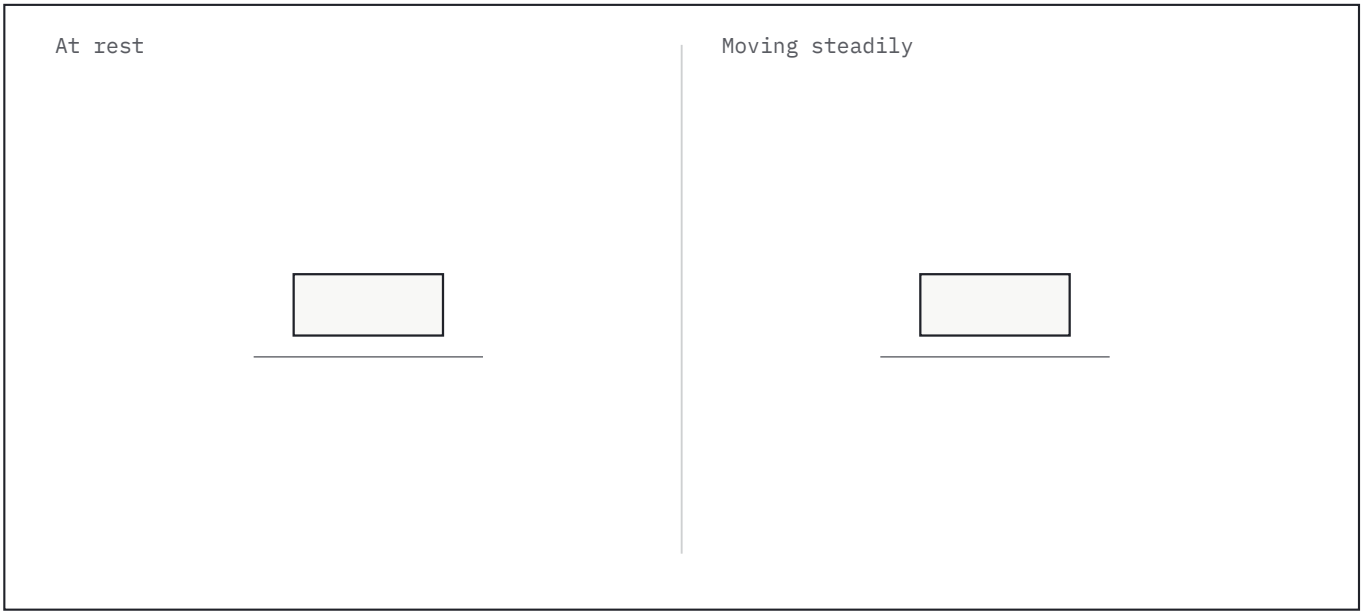
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, equilibrium means …	A cart rolls straight at steady speed. What must the forces on it add up to? The forces add up to ___ because the velocity ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.

Labels you can use: balance · upward part · weight.



Tell a partner: The ___ forces balance. Its acceleration is ___.

On assignment: Equilibrium

SECTION E / BACK / FINAL MEETING ONLY Name _____

The flight display reads “net force zero.” XJ’s last velocity was +3 km/s. Two graphics say STOPPED and COASTING.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Flight controller**

Select the status graphic. Plot a second position two seconds later relative to the first and explain your instruction.

☐ **Broadcast producer**

Choose the lower-third for a live shot. Ask the anchor to distinguish zero acceleration from zero speed before airing it.

Word bank: balance · upward part · weight.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”